

SANJEEVNI

App Feature & UI/UX Design Specification

A Secure, AI-Powered Personal Health Companion for Disaster Resilience

Smart India Hackathon 2026 — Problem Statement 26181 — Team BrajCoders (Team ID 1111)

Purpose of this document. This is a two-part specification. **Part I** lists and prioritizes every feature the app will have, using the 80/20 principle: which 20% of features deliver 80% of the perceived value to a mentor, a judge, and – most importantly – a real Indian user in a heat wave, flood, or pollution event. **Part II** is the developer-facing design document: for every feature, what to build, how it should look and behave (UI/UX), and *why* it is designed that way for the Indian context specifically. Anyone on the team should be able to read Part I alone and explain the entire app to a judge.

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1 Design Philosophy – Read This Before Anything Else

A dashboard that mirrors sensor values (“HR: 72, SpO2: 98%, Temp: 32 deg C”) is not a product – it is a science-fair demo. **The product is the layer of judgment sitting on top of the numbers.** Every screen in Sanjeevni must answer exactly one of three questions for the person looking at it:

1. **“Am I okay right now?”** – status, in plain language, not a number.
2. **“What should I do about it?”** – one clear, small, doable action.
3. **“Who else needs to know?”** – escalation to family, caregiver, or emergency services.

The Indian-context constraints that shape every decision in this document:

- Rural internet penetration is **48.31 per 100** vs. **126.80 per 100** in urban India – the app must be fully useful with zero or intermittent connectivity, not just “work offline as a fallback.”
- The primary at-risk users – elderly citizens and outdoor laborers – are frequently **not** the most digitally literate members of their household. A family member is often the one who will actually monitor the app.
- Health literacy varies widely; numbers (SpO2 94%, heat index 41 deg C) mean very little to most users without translation into an action.
- Multiple Indian languages, and in many cases low reading literacy even in one’s own language, means **voice and icons must be first-class, not accessibility add-ons.**
- Trust in unfamiliar health-tech is reasonably low; the app must be transparent about what it is (a risk-alerting companion) and is *not* (a diagnostic medical device).
- The real distribution and adoption path is institutional (ASHA workers, Primary Health Centres, disaster-response agencies, employers of outdoor labor) as much as individual consumers – the app must serve both a single wearer *and* someone monitoring many people at once.

2 Part I — Complete Feature List (Prioritized, 80/20 Principle)

This section is written so that any team member can read it top to bottom and be able to explain the *entire* app to a mentor or judge in five minutes, without touching code. Every feature is tagged by build priority, following the same scheme as the team’s build guide:

[P1 – CORE DEMO FEATURE]	The demo does not work without this – build first
[P2 – HIGH VALUE]	Makes the demo genuinely convincing and matches judging rubric directly
[P3 – IF TIME PERMITS]	Build only after all P1/P2 are done across the whole team

2.1 PS Requirement 1: Continuous Health Monitoring

- **[P1 – CORE DEMO FEATURE]** Live Vitals screen: heart rate, HRV-derived stress cue, motion state, environment context – shown as **status**, not a raw table.
- **[P1 – CORE DEMO FEATURE]** Personal baseline learning (rolling per-user average of resting HR, RR-variance, heat tolerance) – “normal for you,” not a textbook number.
- **[P2 – HIGH VALUE]** Daily wellness summary card (“Today was a normal day”).
- **[P3 – IF TIME PERMITS]** Sleep quality (inferred from night-time motion + HR patterns, since no dedicated sleep sensor exists yet).

2.2 PS Requirement 2: AI-Based Health Anomaly Detection

- **[P1 – CORE DEMO FEATURE]** Real-time cardiac risk flag (rhythm-anomaly z-score vs. personal baseline, gated by signal quality and motion).
- **[P1 – CORE DEMO FEATURE]** Confidence-aware risk levels (NORMAL / CAUTION / RISK) – never a false sense of precision.
- **[P2 – HIGH VALUE]** “Why am I seeing this?” evidence trail in plain language (translates your engine’s internal `evidence[]` array into a human sentence).
- **[P3 – IF TIME PERMITS]** Trend-based early warning (predicting a rise 15–30 minutes ahead, not just flagging the present).

2.3 PS Requirement 3: Disaster-Specific Health Alerts

- **[P1 – CORE DEMO FEATURE]** Heat-Wave Mode: heat-index-based warning + a personalized “safe hours to work outside today” plan.
- **[P2 – HIGH VALUE]** Air-Quality Mode: AQI-based respiratory-risk alert + best/worst hours for outdoor exposure.
- **[P2 – HIGH VALUE]** Flood/Cyclone Advisory: pulled from public IMD/NDMA feeds for the user’s district, paired with a personal note for vulnerable users.
- **[P1 – CORE DEMO FEATURE]** Every alert type ships with a matching “What to do” action card (symptoms to watch, immediate first steps) – cheap to build, highest value-per-effort feature in the whole app.

2.4 PS Requirement 4: Environmental Awareness

- **[P1 – CORE DEMO FEATURE]** Live environment strip (temperature, humidity, AQI) with source shown (garment sensor vs. public API) – builds trust by being honest about data provenance.
- **[P2 – HIGH VALUE]** Personalized environmental recommendation engine (same AQI reading gives different advice to a healthy adult vs. a vulnerable user).

2.5 PS Requirement 5: Privacy-Preserving Edge AI

- **[P1 – CORE DEMO FEATURE]** Persistent, always-visible “On-Device” / “Nothing leaves your phone” trust indicator.
- **[P1 – CORE DEMO FEATURE]** Plain-language “How this works” screen (what the app does and explicitly does *not* do).
- **[P2 – HIGH VALUE]** Data-sharing control panel (what, if anything, syncs to a caregiver or PHC dashboard, and an explicit opt-in toggle).

2.6 PS Requirement 6: Emergency Assistance Features

- **[P1 – CORE DEMO FEATURE]** One-tap, always-reachable SOS button with visible count-down (cancellable) before auto-send.
- **[P1 – CORE DEMO FEATURE]** Automatic fall/distress detection triggering the same SOS flow.
- **[P1 – CORE DEMO FEATURE]** SMS-based SOS (works with zero data connectivity) with a pre-drafted, informative message and last-known location.

- **[P2 – HIGH VALUE]** Caregiver/Family Companion View – read-only linked account for a family member, who also receives RISK-level alerts directly.

2.7 PS Requirement 7: Personal Wellness Dashboard

- **[P1 – CORE DEMO FEATURE]** Traffic-light home screen status (green/yellow/red) as the single headline element.
- **[P2 – HIGH VALUE]** Category risk scores (cardiac / heat / respiratory / fall) shown as simple gauges, each with one-line plain-language explanation.
- **[P2 – HIGH VALUE]** Weekly trend summary in plain language, not a raw graph.
- **[P3 – IF TIME PERMITS]** Full historical trend graphs for power users / clinicians.

2.8 PS Requirement 8: Scalable Deployment

- **[P2 – HIGH VALUE]** ASHA Worker / PHC Companion Mode: a single list view of everyone in a health worker’s care area, sorted by risk level.
- **[P3 – IF TIME PERMITS]** Multi-device abstraction layer (band/smartwatch form factor support without redesigning the app).

2.9 Cross-Cutting Features (apply to the whole app, not one PS bullet)

- **[P1 – CORE DEMO FEATURE]** **Voice-first regional language support** – spoken alerts, not just translated text; single highest-leverage feature for elderly and low-literacy users.
- **[P1 – CORE DEMO FEATURE]** **Icon-first, low-literacy-friendly onboarding** – tap-a-picture vulnerability profile instead of a medical form.
- **[P2 – HIGH VALUE]** **Visible “Working Offline” badge** – reinforces the core disaster-resilience promise every time the app is opened.
- **[P3 – IF TIME PERMITS]** Accessibility pass (large-text mode, high-contrast mode) for elderly users with vision difficulty.

The 20% that delivers 80% of the value, if the team must cut scope: traffic-light home screen + plain-language “what to do” alert cards + one-tap SOS with SMS fallback + voice alerts in a regional language + the caregiver view. These five features alone turn Sanjeevni from “a wearable that shows numbers” into “a health companion that protects a family,” which is the entire pitch.

3 Codebase Folder Structure (Reference)

```

sanjeevni/
|-- ai-engine/                                (Phase-1 DSP + statistics engine -- DONE)
|   |-- ecg/                                  filters, R-peak, HRV, rhythm anomaly, SQI
|   |-- environment/                          heat index, AQI source, respiratory risk
|   |-- motion/                              activity state, fall detection
|   |-- fusion/                              multi-sensor fusion
|   |-- decision/                            false-alarm gate, risk decision engine
|   |-- baseline/                            personal baseline learning
|   '-- index.ts                             processSensorTick() orchestrator
|
|-- esp32-firmware/
|   '-- sanjeevni_firmware.ino                sensor acquisition + BLE/serial packetizing
|
|-- server/
|   '-- src/
|       |-- serialBridge.ts                  serial parsing, packet -> SensorTickInput
|       '-- server.ts                        HTTP + WebSocket broadcast to frontend
|
'-- app/                                       <-- THIS DOCUMENT DESIGNS THIS LAYER
    |-- src/
    |   |-- screens/
    |   |   |-- OnboardingFlow/              (Sec. 4.1)
    |   |   |-- HomeStatus/                  (Sec. 4.2)
    |   |   |-- DisasterModes/               (Sec. 4.3)
    |   |   |-- EmergencySOS/                (Sec. 4.4)
    |   |   |-- WellnessDashboard/           (Sec. 4.5)
    |   |   |-- CaregiverView/               (Sec. 4.6)
    |   |   |-- TrustAndPrivacy/             (Sec. 4.7)
    |   |   '-- AshaWorkerMode/              (Sec. 4.8)
    |   |-- components/
    |   |   |-- StatusOrb.tsx                traffic-light home indicator
    |   |   |-- ActionCard.tsx               "what to do" card, reused everywhere
    |   |   |-- RiskGauge.tsx                category risk score gauge
    |   |   |-- VoiceAlertPlayer.tsx         TTS playback for regional language
    |   |   |-- OfflineBadge.tsx             connectivity state indicator
    |   |   '-- SOSButton.tsx                global emergency trigger
    |   |-- i18n/                             regional language strings + TTS assets
    |   |-- services/
    |   |   |-- websocketClient.ts           connects to server.ts /ws
    |   |   |-- smsService.ts                native SMS SOS dispatch
    |   |   '-- localStorage.ts              SQLite -- readings, alerts, profile
    |   '-- state/                           app-wide status/risk state store
    '-- assets/
        |-- icons/                           vulnerability-profile picture set
        '-- voice/                           pre-recorded regional-language phrases

```

4 Part II — Developer Design Document (Screen by Screen)

4.1 Onboarding Flow

[P1 – CORE DEMO FEATURE]

What it is. The very first experience – must work for a first-time smartphone user with no assumed literacy.

UI/UX specification.

- Screen speaks its own instructions aloud on load, in the language selected on the very first tap (large flags/scripts, not a dropdown list).
- Vulnerability profile step: a grid of **6–8 large icon cards** (elderly person, outdoor worker, pregnant, heart condition, child, none-of-these) – user taps as many as apply. No text typing required at all for this step.
- Emergency contact step uses the **native contact picker**, never a manual phone-number text field – typing a phone number correctly on a small keyboard is a common failure point for elderly users.
- A single “Calibrating your normal...” screen with a simple animated progress ring while the app collects ~60 seconds of resting baseline data – explained as “We’re learning what’s normal for your body,” never using the word “baseline” or any technical term in-app.

Why this design, for India specifically. Manual form-filling is the single biggest drop-off point for low-literacy and first-time smartphone users. Every step above replaces typing with tapping a picture or using a built-in phone capability (contacts) the user already understands from making calls.

4.2 Home / Status Screen

[P1 – CORE DEMO FEATURE]

What it is. The one screen a user should be able to glance at and understand in under three seconds – this is the screen a judge will remember.

UI/UX specification.

- **Top of screen, largest element: a single colored orb/card** (green / yellow / red) with one short sentence written the way a person would say it out loud:
 - **Green:** “You’re doing fine. Stay hydrated, it’s a hot afternoon.”
 - **Yellow:** “Your heart rate is a bit high for resting. Sit somewhere shaded for a few minutes.”
 - **Red:** “This doesn’t look normal. Please sit down and consider calling for help.”
- Below the status sentence, in smaller/secondary typography: the actual numbers (HR, SpO2, temperature, AQI) for users who want detail – **numbers are secondary information, never the headline.**
- A persistent, small “Working Offline” / “Connected” badge in the header – always visible, never hidden in settings.
- A persistent, small lock-icon “On-device – nothing leaves your phone” badge near the status card.
- The SOS button is always visible as a floating element on this screen, not nested in a menu.

Why this design. A traffic light plus one plain sentence is understandable to *any* literacy level, *any* language (before translation even happens, color + tone of voice already communicate urgency), and any age group. This single design decision is what separates Sanjeevni from a generic vitals-tracking app – it is stated first because it is the highest-leverage screen in the entire product.

4.3 Disaster-Specific Modes

[P1 – CORE DEMO FEATURE] (Heat Wave) [P2 – HIGH VALUE] (Air Quality, Flood/Cyclone)

What it is. Context-aware modes that activate automatically based on environmental thresholds, rather than the user having to seek them out.

UI/UX specification.

- **Auto-activating banner**, not a menu item the user must find – when heat-index/AQI/public advisory thresholds are crossed, a banner appears at the top of the Home screen: “Heat Wave Alert Today.”
- Tapping it opens a single-screen **daily plan**, not a wall of data: “Best hours to work outside today: before 10am, after 5pm.” This reframes a temperature number into a schedule – the single most actionable transformation for an outdoor worker.
- Every disaster mode ships with a matching, ultra-short “**What to do**” **Action Card**: 3–4 bullet points maximum, large text, e.g., symptoms of heat stroke to watch for, what to drink, what to avoid, how to cool someone down while waiting for help.
- Flood/Cyclone Advisory pulls public IMD/NDMA district-level alerts (via public API when online) and layers a one-line personal note for vulnerable-flagged users: “A flood alert is active in your area. If you have a chronic condition, keep your emergency contact informed of your location today.”

Why this design. This is the module that most directly answers the Problem Statement’s explicit call for “resilience during heat waves, floods, pollution events, and other disasters.” It is also the cheapest-to-build, highest-perceived-value feature in the entire app: it requires no new hardware, only good content design and threshold logic already available from `heatIndex.ts` and `respiratoryRisk.ts`.

4.4 Emergency SOS Flow

[P1 – CORE DEMO FEATURE]

What it is. The safety-critical path – must be designed for a person in genuine distress or panic, not a calm browsing session.

UI/UX specification.

- One large, high-contrast button reachable from *every* screen in the app (floating action button, not buried in a tab).
- On automatic fall/distress detection: a full-screen takeover with a large, visible, audible **10–15 second countdown** and one giant “I’m OK, Cancel” button – directly reflects the engine’s own fall-detection state machine (FREEFALL → IMPACT → CONFIRMED), giving the human a chance to override a false positive before anything is sent.
- SOS message is pre-drafted and informative, never just “Help”: “[Name] may need help. Possible [fall/heat stress] detected near [location] at [time]. Sent automatically by Sanjeevni.”
- **SMS is the primary delivery path**, not a fallback – works on any phone, any network quality, with zero data connectivity, matching the app’s core offline-first promise.

- Post-SOS screen shows delivery confirmation (“Message sent to Ramesh (son)”) so the user isn’t left wondering if it worked.

Why this design. Rural network reliability makes SMS – not app notifications or data-based messaging – the only dependable channel during an actual disaster, which is precisely when this feature matters most. The cancellable countdown is essential given the engine’s own honesty about false-positive risk in fall detection.

4.5 Personal Wellness Dashboard

[P2 – HIGH VALUE]

What it is. The screen for someone who wants more than the one-line status – trend and history, still in plain language.

UI/UX specification.

- Four simple gauges (cardiac / heat / respiratory / fall risk), each a simple dial from green to red, each with exactly **one sentence** underneath translating the engine’s **evidence[]** array into plain language (e.g., “Your heart rate has been a little higher than usual today, but nothing urgent.”).
- A daily card: “Today was a normal day. Your resting heart rate stayed close to your usual.” – gives the app a reason to be opened even on days nothing goes wrong, which matters for long-term retention/wear compliance.
- Weekly view in plain language sentences first, with an optional graph underneath for anyone who wants it – never lead with the graph.

Why this design. A pure-alert app that stays silent 95% of the time gets forgotten and eventually stops being worn. A short, positive daily summary builds a habit loop and keeps the device on the body – which is the actual precondition for all the safety features to work at all.

4.6 Caregiver / Family Companion View

[P2 – HIGH VALUE]

What it is. A linked, simplified view for a family member – often the one who is more digitally engaged than the wearer.

UI/UX specification.

- Read-only mirrored version of the Home Status screen for the linked person, plus a small list if monitoring more than one family member (e.g., both parents).
- Direct push notification to the caregiver on any RISK-level alert – **in addition to**, not instead of, notifying the wearer.
- Simple “Last seen active” / “Last check-in” timestamp, so a distant family member has a lightweight way to know their parent is okay without needing a phone call every day.

Why this design. Elderly users are frequently *not* the ones who will check a phone app daily – a child or relative, possibly in a city with better connectivity, will. Routing critical alerts to whoever is most likely to act on them, not just the wearer, is what actually turns a detected risk into a real-world intervention. This is also a genuine differentiator no competitor in the comparison table offers.

4.7 Trust & Privacy Screen

[P1 – CORE DEMO FEATURE]

What it is. A plain-language explanation of what the app is and is not – necessary given reasonable public wariness of unfamiliar health-tech.

UI/UX specification.

- One screen, plain language, no jargon: “This device does NOT diagnose disease. It watches for early warning signs and tells you when something looks unusual for your body.”
- A persistent, small badge visible from the Home screen (not just buried in this settings-like screen) reiterating “Nothing leaves your phone.”
- A simple data-sharing control: toggle for whether anything syncs to a caregiver or, in institutional deployments, a PHC/ASHA worker dashboard – explicit opt-in, never default-on.

Why this design. Builds real trust rather than assuming users will read a privacy policy (they won’t). Also protects the product from over-reliance/misuse claims by being explicit about its non-diagnostic scope directly in the product, not just in the pitch deck.

4.8 ASHA Worker / PHC Companion Mode

[P2 – HIGH VALUE]

What it is. A separate, simple mode for a health worker monitoring many people at once – this is the feature that turns Sanjeevni from a single-user gadget into public health infrastructure.

UI/UX specification.

- A single list view of everyone in the worker’s care area, **sorted by risk level** (red first), each row showing name, one-line status, and last-updated time.
- Tapping a person opens their simplified Home Status view (read-only) – no need to learn a second interface.
- Designed for quick daily scanning on a basic Android phone, not a data-dense clinical dashboard.

Why this design. This directly matches the actual go-to-market and distribution path identified for reaching rural populations at scale (ASHA/PHC networks), and lets one health worker efficiently prioritize which households to visit first during a heat wave or flood – a capability no competitor product offers, and one that directly strengthens the Innovation and Potential Impact scoring criteria.

4.9 Voice & Regional Language Layer (Cross-Cutting)

[P1 – CORE DEMO FEATURE]

What it is. Not a single screen – a capability threaded through every alert and status message in the app.

UI/UX specification.

- Every status sentence and disaster alert has a matching spoken version, triggered automatically for RISK-level alerts (not just available on request) – e.g., a spoken regional-language line such as “*Bahut garmi hai, paani piyen*” (Hindi: “It’s very hot, drink water”) reaches a low-literacy user that a written banner alone will not.
- Language selection happens once at onboarding via icons/scripts, and every subsequent alert – text and voice – respects it without the user needing to configure anything again.
- Icon-first navigation as a fallback throughout the app, so core actions (SOS, “I’m OK,” view status) never depend on reading any language at all.

Why this design. This is the single highest-leverage feature for the app’s actual target users (elderly, rural, potentially low-literacy) and is explicitly what none of the competitor products in the comparison table offer. It costs comparatively little engineering effort (text-to-speech + translated string tables) relative to how much it changes real-world usability – a textbook 80/20 feature.

5 Summary: What to Build First (Team Priority Order)

Build in this order for maximum demo impact per hour spent:

1. Home Status screen with traffic-light orb + plain-language sentence (Sec. 4.2)
2. SOS button + cancellable countdown + SMS dispatch (Sec. 4.4)
3. Heat-Wave Mode banner + daily-plan card + What-to-do action card (Sec. 4.3)
4. Onboarding with icon-based vulnerability profile + contact picker (Sec. 4.1)
5. Trust & Privacy screen with persistent “on-device” badge (Sec. 4.7)
6. Voice alert playback for at least one regional language (Sec. 4.9)
7. Caregiver View (Sec. 4.6) and ASHA Worker Mode (Sec. 4.8) as stretch/P2, once everything above is demo-ready

This order intentionally puts *judgment and communication* features ahead of *more dashboards* – because that is exactly the gap between a hackathon vitals-tracker and a product a real Indian household would actually trust and use.